

The general power-step tower:

$$Z_d(c) = \left(\prod_{i < d} k_i^{-1} \right) D_d c^{-q_d-1} G_d(cb^{S_d}) \text{ for every } (k, h)$$

Claude, with Daniel Murfet

2026-09-06

Abstract

All exact reductions so far (units 44, 50, 52) are instances of one recursion. With $q_i = (h_i + 1)/k_i$ and $q_{-1} = 0$, define $G_0 = f$, $G_{j+1}(y) = \int_0^y x^{q_j - q_{j-1} - 1} G_j(x) dx$ and $D_0 = 1$, $D_{d+1} = D_d (b^{S_d})^{q_{d-1} - q_d}$ with $S_d = \sum_{i < d} k_i$. Then for every d and every exponent vector, $Z_d(c) = (\prod_{i < d} k_i^{-1}) D_d c^{-q_d-1} G_d(cb^{S_d})$. The all-equal tower is $\text{iterDivPrim}(F_{p-1})$ and the second level with exponents (q, p) is unit 52's noncritical base. The formula was checked by three-dimensional quadrature with three distinct exponents to $5 \cdot 10^{-14}$. Zero `sorry/axiom`.

1 Setting

Each coordinate contributes a chart power substitution and a scaling; the only data that changes from level to level is the exponent q_d of the new coordinate versus the exponent q_{d-1} carried by the previous level, which appear as the power $x^{q_d - q_{d-1} - 1}$ in the next integrand and as the factor $B^{q_{d-1} - q_d}$ in the scale. The three earlier reductions correspond to $q_d = q_{d-1}$ (power x^{-1} : a logarithm), $q_d < q_{d-1}$ (a convergent integral) and the base level $q_{-1} = 0$ (the weighted primitive).

2 The things formalised

```

noncomputable def powStep (s : ℕ) (G : ℝ → ℝ) (y : ℝ) : ℝ := x in Ioc 0 y, x ^ s * G x
noncomputable def genTower (a : ℝ) (k h : ℕ → ℕ) : ℝ → ℝ
| 0 => quadKernel a
| j + 1 => powStep (chartExp k h j - prevExp k h j - 1) (genTower a k h j)
theorem iterChartGen_step_general ... (hZ : ℝ < 0, 0 < c →
  iterChartGen a b k h d c = K * c ^ (-q') * D * G (c * B)) (c : ℝ) (hc : 0 < c) :
  iterChartGen a b k h (d + 1) c
  = K * (1 / (k d : ℝ)) * c ^ (-(chartExp k h d)) * (D * B ^ (q' - chartExp k h d))
  * powStep (chartExp k h d - q' - 1) G (c * (B * b ^ (k d)))
theorem iterChartGen_eq_genTower ... (d : ℕ) (c : ℝ) (hc : 0 < c) :
  iterChartGen a b k h d c
  = (i Finset.range d, (1 : ℝ) / k i) * genScale b k h d * c ^ (-(prevExp k h d))
  * genTower a k h d (c * b ^ (i Finset.range d, k i))
theorem genTower_eq_iterDivPrim ... :
  genTower a k h (d + 1) = iterDivPrim (weightedPrimitive a (p - 1)) d
theorem genTower_two_eq_noLogBase ... : genTower a k h 2 = noLogBase a p q

```

3 Proof strategy

The general step is unit 52's step lemma with the exponent of the previous level freed: after the chart power substitution the integrand is $s^{q-1}s^{-q'}G(cBs) = s^{q-q'-1}G(cBs)$, the scaling lemma with $\gamma = q - q' - 1$ produces $(cB)^{-(q-q')}$, and $c^{-q'}c^{-(q-q')} = c^{-q}$ leaves $B^{q'-q}$ as the new scale factor. The exact reduction is an induction on d generalising c ; the base case $d = 0$ is trivial because all conventions ($q_{-1} = 0$, $D_0 = 1$, $S_0 = 0$) make $Z_0 = f$ fit the general shape. The two identifications are inductions in which the power x^{-1} is rewritten as division and the definitions unfold by **rfl**.

4 Roadblocks and resolutions

- The final goal of the induction step differed only by the order of factors; **ring**.
- Unfolding **weightedPrimitive** under a binder with **rw** fails; the goal was closed by **rfl** since all three definitions unfold definitionally.

5 What was Mathlib, what was new

Mathlib: `Real.mul_rpow`, `Real.rpow_add`, `Finset.prod_range_succ`, `Finset.sum_range_succ`.
New: the general tower, its exact reduction for arbitrary (k, h) , and the identifications with the earlier special towers.

6 Lessons

The right generality for an exact reduction is the one in which the base case is trivial: with $q_{-1} = 0$ and $D_0 = 1$ the kernel itself is a tower of height zero, and no special treatment of the innermost coordinate is needed.

7 Outcome

The chart standard integral with constant ξ is now reduced exactly, for every exponent vector, to a one-dimensional tower G_d . The remaining analytic problem is the growth of $G_d(y)$ as $y \rightarrow \infty$, expected to be $y^{q_{d-1}-q_{\min}}(\log y)^{m-1}$ with m the multiplicity of the minimal exponent; this would give the full “only minimal exponents count” theorem.